

STATEMENT OF OBJECTIVES

Optical Radiation Bioeffects and Safety (ORBS)

As of 09 December 2025

Controlled By: AFRL/RHDO

CUI Category: OPSEC

Dissemination Control: None

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Background, Goal, Scope, Objectives

1.0 Background: The Air Force Research Laboratory's (AFRL) Optical Radiation Bioeffects Branch (RHDO) is a key component of the Bioeffects Division (RHD) under the Human Effectiveness Directorate (RH) within the 711th Human Performance Wing (711 HPW). The branch comprises a diverse group of scientists and engineers whose mission is to understand the short and long-term biological effects associated with Directed Energy (DE) technologies. Research activities primarily focus on technologies spanning ultraviolet through near-infrared wavelengths of the electromagnetic spectrum, however, other forms of DE exposures (e.g., acoustics, ultrasound, particle beam, etc.) may be investigated as required by United States Air Force (USAF) and Department of Defense (DoD) demands.

2.0 Goal: The Optical Radiation Bioeffects and Safety (ORBS) program conducts research studies with DE-based devices and systems to deliver accurate, evidence-based information to DoD and industry stakeholders, national and international health and safety standards committees, and the public. The goal is to enable optimal use and maximum safe exploitation of the electromagnetic spectrum for national defense by protecting personnel and communities and evaluating the applications of DE systems.

3.0 Scope: This Statement of Objectives (SOO) describes the work that shall be performed by the Contractor to support the 711 HPW/RHDO in a full range of research and experimentation efforts as described in the Continuing Human Enabling, Enhancing, Restoring, and Sustaining (CHEERS) Multiple Authority Announcement (MAA) (# FA2384-24-S-2233) SOO (attachment 2) RHD Technical Requirements (section 3.2). Research activities under this program are anticipated to follow the RH organizational construct as described in CHEERS, i.e., research will be categorized based on the technology readiness level (TRL). The Contractor should expect to perform basic to early applied research and advanced technology development and transition. The basic to early applied research will align with the Bioeffects Core Technical Competency (CTC), and the advanced technology development and transition activities will align with the Bioeffects Product Line (PL). For further details on CHEERS and/or the RH organizational construct, please review the CHEERS MAA notice and CHEERS MAA Technical Requirements SOO on SAM.gov.

Lasers are essential for range finding, precision delivery of munitions, target illumination when combined with night-vision goggle use and are being developed for advanced applications such as missile defeat. Lasers can pose threats to missions or can enhance capabilities for security forces and special operations by visual jamming. To address risks to aircrews, the Contractor shall expand methods to: minimize injury to aircrews; contribute to the establishment of occupational safety standards and flight tactics, techniques, and procedures (TTPs); define specifications for countermeasures; transition laser eye protection (LEP); enhance safety at ranges; develop procedures for laser injury assessment; and conduct research to determine appropriate technologies for non-lethal weapons (NLW). USAF and joint service payoffs include validation of effectiveness of non-lethal optical technologies, safe use of ranges, compliance with SECDEF policy on laser weapons, data for Electronic Information Security processes, mission planning for laser threats, and LEP acceptable for combat operations.

ORBS technical objectives are categorized under three main functional areas for which technical expertise will be required:

- 1) Novel Laser Bioeffects
- 2) Nuclear Flash and LEP
- 3) DE Modeling, Simulation, and Analysis

Within these functional areas, the Contractor shall characterize the fundamental bioeffects associated with DE technologies, optimize/assess the safety and effectiveness of DE systems, develop and assess dosimetry tools, provide modeling and simulation support to laboratory experiments and DE applications, design and characterize protective devices, investigate DE countermeasures, and provide science-based information to the DoD to support national and international safety standards.

4.0 Objectives: The three functional areas and their respective technical objectives are intended to steer the research and development efforts of this program. These areas are also designed to address critical research challenges, advance technological capabilities, and achieve measurable outcomes that support the overarching mission of RHDO. It is expected that across the three functional areas, research efforts will span from the lowest level of biological interaction, i.e., molecular, through the highest level of interaction, i.e., whole organisms. A non-exclusive list of knowledge, skills, and abilities to successfully execute RHDO's mission can be found at the end of this section in Table 1.

Offerors may propose on all of, any of, or a niche within these three broad research areas. In any of the instances where the research/experimentation described references any of various international agreements or entities, applicable export control statutes, to include requirements for licenses, will apply.

4.1 Novel Laser Bioeffects: Plan and execute high-quality empirical research on the bioeffects of DE laser exposures.

The Contractor shall conduct laboratory and/or field studies to determine the short and long-term bioeffects associated with laser exposures, biological damage mechanisms and thresholds, and laser-tissue interactions for given optical parameters; and shall determine the effectiveness of using optical exposures to produce any militarily relevant behavioral response in humans. In addition, the Contractor shall perform research investigating fundamental visual mechanisms such as color vision, visual acuity, photostress recovery, and visual perception to assess the mission impact of bright light to humans; and develop advanced protection technologies, low-vision models and simulations, threat analysis systems, and special test equipment that measure advanced optical technologies and their countermeasures. Collaborative basic research with US and non-US universities, other government organizations may be anticipated, with potential to promote and expand this area of research. Collaborative research with medical partners may also be anticipated, with potential focus on diagnostic/triage technologies and medical treatment strategies. Technical Objectives are listed below:

- a. Operate and maintain the laser systems required to accomplish the research objectives. The Contractor shall operate and maintain numerous laser systems to include but not limited to high energy laser systems (HEL), existing wavelength-tunable high pulse power titanium-Sapphire laser, femtosecond laser systems, multiple high-energy continuous wave (CW) lasers, and tunable nanosecond optical parametric oscillators,
- b. Plan and conduct experiments advancing state-of-the-art research and development in biophotonics and its technology applications. Plan and conduct experiments to determine probabilities and thresholds of ocular or skin damage and damage mechanisms from both acute and long-term chronic exposure to ultraviolet, visible, and infrared lasers. The experiments shall include studies at the *in-vitro* and *ex-vivo* levels, as well as at the level of both animals and humans (*in-vivo*), with supporting

computer modeling at all levels. The assessment shall include using advanced imaging techniques for experiments at all three of these levels to determine or extrapolate mission impact on human subjects through post-event evaluation in support of RHDO and customer funded goals.

- c. Incorporate the findings from advanced laser-tissue interaction research, including damage thresholds and mechanisms, for validation and possible contribution to the national and international health and safety standards.
- d. In coordination with the Integrated Product Team, develop test plans and experiments using these lasers, to expand the knowledge of bioeffects specific to their frequency and pulse width domains. Develop and execute experimental plans, test plans, operate and maintain numerous laser systems (including advanced CW through pulsed femtosecond laser systems) and related test equipment.
- e. Assist in the development of tissue and animal use protocols as required to support bioeffects research efforts involving tissue and animals. Experiments should be designed to maximize the potential to extrapolate results and data from animal subjects to humans.
- f. Design, develop, and evaluate hardware to exploit personnel susceptibilities based on bioeffects modeling.
- g. Perform in-house and on-field HEL-target reflection and bidirectional reflection distribution function (BRDF) measurements to validate and improve laser hazard prediction models and establish/maintain database of measured BRDFs.

4.2 Nuclear Flash and LEP: Plan and execute high-quality research on human effects of LEP characterization and performance.

The Contractor shall design and conduct experiments for human effects on performance and LEP characterization and measurements to test and evaluate LEP to counter emerging DE weapons with developing technologies for agile wavelength threats. The Contractor shall perform augmented reality (AR)/virtual reality (VR) laser dazzle experiments to validate and integrate into models for LEP development and wargaming scenarios. The Contractor shall perform field studies to validate laboratory data to provide high-quality protective equipment to the warfighter. The contractor support shall include measuring and modeling performance of demonstration eyewear, tests of eyewear optical quality and physical properties, and laboratory psychophysical vision tests with human subjects to assess the human performance impacts of this eyewear. The Contractor shall perform simulator, ground, and flight tests of the laser eye protection devices, and develop new methods to assess operational effectiveness, develop 3D eyewear models in a software design environment, integrate emerging eyewear designs, and perform on-site prototyping of LEP devices. Technical objectives are listed below:

- a. Research, characterize, and evaluate eye protection to counter emerging directed energy weapons, laser systems with advanced technologies for agile laser wavelengths, and nuclear flash.
- b. Develop techniques for computer modeling of current protective systems, multiple aircrew helmets, oxygen masks, head forms and other life support equipment.
- c. The Contractor shall extend the scientific and engineering databases to enhance capabilities to measure and assess protective measures. These databases include optical properties and physical properties of the protective systems up to and including all necessary parametric data on life support protective equipment. This database also includes performance factors so that it can become a basis

for detailed predictive models of human performance under various operational conditions, with and without the protective systems.

d. Develop a vulnerability model using computer modeling techniques, enabling future validation of newly designed protective systems. Develop optical models predicting personnel susceptibility to optical glare sources. Integrate bioeffects and vision models into a single tool to aid development of eye protection devices and transition the tool to acquisition partners.

e. Perform field evaluation studies of eye protection devices to validate laboratory data and modeling results. Performance evaluations shall be conducted on a non-interference basis under realistic operational conditions to include weapon system trainers, operational aircraft, and other battlefield training situations. The Contractor shall determine, through field experimentation, the human factors and systems compatibility issues between developmental eye protection systems and the aircraft and other operational systems in which the users must function. Human use experimental protocols shall be developed as necessary for field testing.

f. Assist in the development of human use protocols investigating impact of lasers and LEP on human effectiveness, task performance, and personnel susceptibility and vulnerability to single and combined laser sources and threats.

4.3 Modeling, Simulation & Analysis (MS&A): Enhance existing and develop new MS&A tools and software products related to fundamental laser-tissue interactions, engineering, engagement, and mission level models.

The Contractor shall develop and enhance MS&A tools to mature advanced theoretical and numerical models describing fundamental laser-tissue interactions to include energy propagation, thermal response, thermo-elastic response, damage mechanisms and other multi-physics responses. These numerical tools should inform dosimetric metrics and simulate laser bioeffect experimental findings to demonstrate biophysical understanding. The Contractor shall also validate standards through models of biological responses provided from experimentation. The Contractor shall refine models for LEP development and AR/VR into distributed simulation, and other experimental wargaming scenarios. Software development will include the design and improvement of multi-physics models relevant to the bioeffects of DE exposures across a broad parameter space to include ultrafast lasers, high peak power laser systems, CW, and tunable (e.g., wavelength, pulse repetition rate, and power) laser systems. Models should include the development of safety analysis tools which are capable of predicting DE hazards in multiple simulation environments. This development shall include deterministic HEL range safety, based on the ANSI Z136.1 standard, as well as probabilistic risk methods delineating severity and probability of hazard. The Contractor shall also perform research on the propagation and interaction of DE and kinetic energy (KE) to predict exposure thresholds, through computational physics models that integrate existing thermodynamics and electrodynamics methods. The Contractor shall develop and deliver a software development plan for mature software models prior to implementation. This plan should include model-based systems engineering (MBSE), traceability of system goals, key functional features, critical stakeholders, system architecture design, and a test plan for verifying correctness of functional capabilities, to include potential verification and validation of the software. The architecture must be at a level of detail sufficient to show functional flow and identify data interface requirements between system functional components and verify ability to integrate components into larger scale products. This plan must enable planning for future modification of the system and include justification for design decisions. Each of the above modeling, simulation, and analysis tools may fall into one or more of the modeling

levels: Physics, Engineering, Engagement, and Mission. These categories define a specific scope of fidelity and intended use of the system. Technical Objectives are listed below:

a. **Physics-Level Modeling:** Employ standard scientific computational languages, physics-level modeling to include the development of first-principal theories, mechanistic models and advanced computational approaches which describe laser-tissue interactions, laser-material interactions, molecular dynamics, cell membrane dynamics, and the evaluation of physiological response factors. The Contractor shall have the ability to prototype/modify models and methodologies using a range of scientific computing languages (e.g., C++, MATLAB, Python, FORTRAN, R, etc.) to develop new capabilities and proof-of-concept analysis. The Contractor shall implement into higher-level languages and languages optimized for high-performance computing (HPC) architectures. As part of the HPC support, the Contractor shall also maintain and administer high-performance workstations and small-scale HPC systems running Linux-based operating system. Leveraging experimental data, these high-fidelity numerical and analytical models integrate the latest data available on dynamic tissue properties and interaction modes to produce a multi-physics level model of tissue response to DE exposure to further inform highly dynamic bioeffects systems. Analytical and semi-analytical solutions to the complex dynamics of laser-tissue interactions should be pursued as a fundamentally sound alternative to more computationally expensive numerical approaches. Analysis is a component of this modeling, requiring the creation of specialized data analysis techniques and synthesis for outputs from large-scale data sets.

b. **Engineering-Level Integration of Models:** Encapsulation of near-real-time algorithms in standard C++ and other common production code languages. This shall include the translation of codes from a variety of individual prototypes into standard production code languages. The activities require the skills for integration of models in a variety of environments (Linux, MS Windows, etc.) and the ability to re-write complex mathematical representations in standard production languages. The Contractor shall also maintain and administer in-house high-performance computing resources running Linux-based operating system. Specific engineering-level modeling efforts include research and characterization of HEL effectiveness by developing new laser safety procedures and safety analysis tools to predict and analyze hazardous zones for HEL and (potentially) kinetic systems. The engineering-level simulation codes shall also include the implementation of radiometric principles from light propagation and scatter, to estimates of dosimetry. These types of models include range safety tools and engagement analysis tools (e.g. IWEA) and are run as desktop applications.

c. **Engagement-Level Models:** Engagement-level modeling includes integrating engineering level models into various simulation architectures (e.g. Endgame Framework, end-to-end frameworks, weaponeering tools, etc.) for engagement-level assessment of the effects of lasers and broadband light on the battlefield. These architectures focus on shooter-to-target and primary and secondary collateral effects engagement scenarios. This type of modeling may include DE/KE hazard model integration into high fidelity weapon analyses for operational risk calculus. These types of tools may be desktop applications, or applications that run in a distributed environment.

d. **Mission-Level Models:** The Contractor shall integrate engineering and engagement-level models into various established mission-level simulation environments (e.g. Advanced Framework for Simulation, Integration, and Modeling (AFSIM), distributed simulations, etc.) for use in mission-level studies of the personnel effects and risks of lasers and broad-band light on the battlefield. The Contractor will support transitions from AFRL through integration of new analysis models and tools to customer platforms. These architectures focus on many-versus-many entity simulation scenarios. This type of modeling may include DE/KE hazard model integration into operator-in-the-

loop battle management systems. These types of tools may be desktop applications, or component utilities that run in a distributed environment.

e. Optical Radiation Safety Desktop Analysis Tools: The Contractor shall provide analysis and analysis tools in coordination with the 711 HPW/RHDO Optical Radiation and HEL Safety team to certify the safe use of optical systems on test and training ranges throughout the Air Force. The Contractor shall help develop, mature, validate, and apply modeling software and analysis to perform laser hazard evaluations and HEL weapon collateral effects modeling that are used by policy-making bodies to formulate safety recommendations based on current ANSI standards (ANSI Z 136.1), AF regulations (AFI 48-139), DoD regulations (DODI 6055.15, CJCSM 3230.01), and ACGIH policy, and shall develop and maintain the research information database, analyze the contents of the database, and develop methods of improving the efficiency of responding to information requests. Conduct research in the instrumentation of outdoor range tests for the purpose of assessing collateral hazards associated with the use of these HEL Initiative Systems. Develop test plans and safety plans for tests.

f. Development, Security, and Operations (DevSecOps): The Contractor shall fulfill proposed software development lifecycle methodology, emphasizing DevSecOps principles, continuous integration/continuous delivery (CI/CD), robust version control, automated testing strategies, and security-by-design practices. It must also address the use of modern development tools and environments to ensure maintainability, scalability, and secure deployment across various platforms (desktop, distributed, cloud). The Contractor is expected to utilize structured, FAIR (Findable, Accessible, Interoperable, Reusable) data management practices, potentially using cloud-native data platforms, and robust metadata standards to facilitate data sharing and long-term archival. The Contractor shall explore and, where appropriate, integrate advanced computational methods, including Artificial Intelligence (AI) and Machine Learning (ML) techniques to enhance model predictive capabilities, optimize simulation performance, and accelerate data analysis.

g. Non-lethal weapons (NLW)/scalable effects weapons (SEW): The Contractor shall conduct simulation and analysis research to determine appropriate technologies for NLW and SEW applications. The Contractor shall conduct, and document efforts involving human effects analyses on candidate NLW and SEW systems, and risk assessments on candidate systems. The Contractors shall develop and/or modify models to characterize the physiological and psychological effects of NLW and SEW payloads, and support field test planning and data interpretation for NLW and SEW systems.

Table 1: Non-exclusive list of knowledge, skills, and abilities by Functional Area to successfully execute RHDO's mission.

Functional Area 1: Novel Laser Bioeffects	
Knowledge	Skills & Abilities

Laser Systems, Supporting Equipment, & Fabrication	Wavelength-tunable high pulse power lasers, ultrafast laser systems, high-energy continuous wave lasers, tunable nanosecond optical parametric oscillators, benchtop lasers, THz laser, optical design & alignment, engineering controls, laser propagation, power meters, calorimeters, infrared cameras, thermal cameras, high-speed cameras, fiber optics, oscilloscopes, network analyzers, delay generators, dosimetry, beam diameter profiling, LabView, MATLAB/Simulink, Python, validation and calibration procedures, laser safety & regulatory knowledge, CAD & prototyping, 3D printing, machining, soldering
Cell Culture	The preparation, propagation & maintenance of primary, immortalized, stem, self-propagating eukaryotic/prokaryotic cell lines/strains under various culture environments Examples include 1) cell lines from ATCC, 2D cell culture, tissues from MatTek; cell media fabrication, & cell-based assays & testing 2) Osmolarity, pH, conductivity; cryo-preservation of cell stocks; immunological & biochemical assays & staining 3) Analysis of cell health/viability, proliferation, apoptosis, cell cycle & other biochemical and cell-based assays
Molecular Biological Assays	DNA/RNA extraction, purification, & quantification methods Examples include 1) Commercial extraction/purification kits, NanoDrop spectrophotometry, gel electrophoresis, DNA/RNA analysis methods 2) Non-radioactive southern & northern blotting, quantitative real-time polymerase chain reaction (QRT-PCR); Recombinant DNA methods 3) Polymerase chain reaction (PCR), Restriction enzymes, bacterial & viral cloning; Epigenetic regulation methods 4) DNA methylation, histone modification, chromatin compaction, & nuclear organization; sequencing 5) Microarray platforms for whole genome sequencing & whole exome sequencing; microarray analysis; RNA interference
Immunology & Immunoassays	Analysis of immune cell functions, cytokine expression, innate & adaptive responses, & cellular/humoral immunity. Examples include cell activation; fluorescent immunoassays; enzyme-linked immunosorbent assays (ELISA); SDS-PAGE and Western blots; immunocytochemistry; fluorescent immunolabeling, immunoprecipitation; flow immunoassay
Microscopy & Spectroscopy	Use of basic microscopy systems to observe/image cells & tissues. Examples include 1) Brightfield, phase contrast & dark field; use of advanced microscopy systems to observe/image molecular responses in real-time 2) Fluorescent, 1-photon confocal, 2-photon confocal, & non-linear optical systems; use of custom-engineered microscopy systems to observe/image cells & tissues; use of microscopy associated software to process and analyze micrographs 3) Zen Blue, Zen Black, Image J, FIJI. Use of basic and advanced spectroscopy systems such as ultraviolet-visible, Fourier Transform Infrared, and Transient Absorption Spectroscopy
Tissue Processing	Compliance with American Association for Laboratory Animal Science/Occupational Health, development of Standard Operating Procedures, <i>ex vivo</i> tissue studies, prepping tissue for histology, hazardous chemical handling, subject handling, familiarity with fundus imaging/slit lamp biomicroscopy/optical coherence tomography/autorefraction, Spatial Frequency Domain Imaging, use of custom-engineered software to map exposures on retinal imaging modalities

Documentation	Tracking laser logs/inventory, tracking laser maintenance, technical report writing, and laboratory notebook/documentation of experimental and optical designs
Functional Area 2: Nuclear Flash & LEP	
Knowledge	Skills & Abilities
Vision Psychophysics	Visual acuity measurements, color perception visual tasks, contrast sensitivity testing, spatial frequency testing, reaction time testing, vision recovery testing, ambient luminant effects on vision
Laser Interaction with Vision	Laser interaction with the eye, optometry principles, eye safety with lasers, optical alignment, visual sensitivity, scotoma, broadband interaction with the eye
Light Filters for Vision	Color discrimination with phosphorescence, Photonic Luminance Transmission/Optical Density Analysis, spectroscopy for material characterization, colorimeters, field of view testing, filters for wavelengths
Human Use Experiments	Human use protocol development, surveys, data safety, experimental design, informed consent, human use data collection
Engineering Principles	Robotic movement, model replicas, electrical circuitry, static and dynamic movement, programming, machining
Functional Area 3: MS&A	
Knowledge	Skills & Abilities
Programming Languages	Languages for core library development (C++, C#), other languages for scientific computing (MATLAB, Python, R, Mathematica, Fortran)
Numerical Analysis	Numerical approaches for continuum mechanics modeling (Finite Difference Methods, Finite Element Method, Finite Volume, spectral), numerical approaches for systems of ordinary differential equations, Boundary conditions, probability and statistics, regression testing, applications of supervised learning approaches, neural networks for deep learning
Software Development	Agile development, Model-Based System Engineering, Unit Testing Frameworks, Version control systems using git, C++ design patterns, database management
Data Visualization & Processing	Python, gnuplot, ParaView, Excel, Computer Aided Design (CAD) software, other visualization/plotting libraries specific to scientific computing libraries
High Performance Computing	Parallel computing using OpenMP & Message Passing Interface, distributed computing, common HPC schedulers torque and Simple Linux Utility for Resource management (slurm), Bash shell scripting, General-Purpose computing on graphics processing units acceleration)
Statistical Analysis	Statistical analysis using R, Python, or other preferred statistical software that perform common descriptive & inferential statistics testing and analysis, Probit analysis, Analysis of Variance

Analytical Methods	Sensitivity analysis, uncertainty quantification, regression testing, Calculus, Linear Algebra, Differential Equations
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Requirements and Other Information

1.0 Human Use Research Protocols; Development and Approval: USAF human use research involves the study of human subjects to advance the understanding of factors that impact the performance, health, and safety of USAF personnel. This research is essential for developing technologies, protocols, and interventions that enhance operational effectiveness and safeguard the well-being of airmen. It demands a high level of expertise to ensure ethical conduct, scientific integrity, and the protection of participants. The use of human subjects in Air Force-funded research is subject to participant rights, ensuring informed consent, and adhering to rigorous regulatory and ethical standards. Hence, the Contractor shall assist in the preparation of research protocols for all human use experiments which shall be prepared well in advance of need, to allow time for review/approval by the 711 HPW Institutional Review Board (IRB). The Contractor shall maintain records associated with human subject research as required by the approved protocol. All such work shall be conducted per the directives and instructions listed below:

- a. Department of Defense, Title 32, Code of Federal Regulations, Part 219 (32 CFR 219), “Protection of Human Subjects” (July 2004);
- b. DHHS Regulations, Title 45, Code of Federal Regulations, Part 46 (45 CFR 46), “Protection of Human Subjects” (Oct 2004);
- c. DoD Directive 3216.2, “Protection of Human Subjects and Adherence to Ethical Standards in DoD-Supported Research,” 25 March 2002;
- d. Air Force Instruction 40-402, “Protection of Human Subjects in Biomedical and Behavioral Research” (Sep 2000);
- e. AFRL Instruction 40-402, “Using Human Subjects in Research, Development, Test and Evaluation” (Feb 1999).

Human testing shall not commence until approval by the 711 HPW IRB has been obtained.

2.0 Animal Use Research Protocols; Development and Approval: USAF animal use research plays a critical role in advancing the safety, health, and operational effectiveness of USAF personnel by providing valuable insights that cannot be obtained through other means. It requires specialized skills and expertise to ensure ethical, accurate, and scientifically valid outcomes. The use of animal subjects in USAF-funded research is subject to humane practices, rigorous methodology, adherence to regulatory standards, and follows the reduce, refine, and replace philosophy. Hence, the Contractor shall assist in the preparation of animal use research protocols which shall be prepared well in advance of need, to allow time for review/approval by the Tri-Service Research Laboratory Institutional Animal Care and Use Committee (IACUC). The Contractor shall maintain records associated with animal subject research as required by the approval protocol. All such work shall be conducted per the directives and instructions listed below:

- a. PHS Policy on Humane Care and Use of Laboratory Animals(10/00);

- b. Federal Animal Welfare Act. Public Law 89-544, 1966 as amended, and it's implementing regulations;
- c. Guide for the Care and Use of Laboratory Animals. National Research Council, published by the Academy Press, revised 1996, or succeeding editions;
- d. DoD Directive 3216.1, Use of Laboratory Animals in DoD Programs, dated 17 April 1995;
- e. AFJI 40-401, dated June 84, The Use of Animals in DoD Programs;
- f. AFR 169-2, Use of Animals in DoD Programs, and AFRL Sup 1, dated 1 June 1984;
- g. Branch Operating Instruction 61-29, Herpes B-virus (Herpes simian) Guidelines.

Animal testing shall not commence until approval by the TSRL IACUC has been obtained.

3.0 711 HPW Biosafety Program: All research under this program shall conform to the applicable biosafety policies, regulations, and guidelines and follow the 711 HPW Operating Instruction 90-800 which establishes the biosafety requirements to use, handle, transport, transfer, store, or dispose of infectious agents and toxins (IAT), recombinant and synthetic nucleic acids, blood, and other potentially infectious materials (OPIM) and establishes the Institutional Biosafety Committee (IBC) as the reviewing and oversight body of the biosafety program. The Contractor shall assist in the preparation of IBC research protocols which shall be prepared well in advance of need, to allow time for review/approval by the 711 HPW IBC. All such work shall be conducted per the directives and instructions listed below:

- a. US Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern, dated 24 September 2015
- b. 711 HPW OI 90-800, 711 HPW Biosafety Program, dated 15 March 2022
- c. 711 HPW PAM 90-800, Biosafety Manual, dated 15 November 2022
- d. DoD Manual 6055.18, Safety Standards for Microbiological and Biomedical Laboratories, dated 11 August 2020
- e. NIH Guidelines for Research Involving Recombinant or Synthetic Nucleic Acid Molecules (NIH Guidelines), dated April 2024

Research activities under the purview of the Biosafety Program and IBC shall not commence until approval by the 711 HPW IBC has been obtained.

4.0 Safety Standards: The Contractor shall support, as directed, the development of safety standards as a participant in national and international technical working groups. The work shall include the development of safety analysis and measurement procedures. Safety standard documents shall be reviewed for accuracy. Example analyses shall be developed for inclusion in standards. Peer-reviewed papers summarizing new methods and research findings shall be authored for publication in support of new safety standard issues. The Contractor shall also support USAF laser safety training and lecture activities through the procurement and development of training materials such as safety standards and briefings.

5.0 Base Support: The Government may provide laboratory space and all necessary equipment for completion of the technical effort. Desk space to accommodate personnel shall be provided to complete necessary planning, analysis, and writing of technical reports to fulfill objectives of the effort. The Government, as a part of its Base support, shall include an office workspace equipped with appropriate computer system(s) and software, facsimile machine, and telephone equipment. The Contractor shall require local access to the Department of Defense computer networks and systems and shall be issued a Common Access Card (CAC).

6.0 Training: The Contractor will be required to participate in the government's in-house and web-based security training program under the terms of the contract. The contractor shall ensure all employees complete Environmental Health and Safety, and Security training in a timely manner; as well as all other special training to include Hazardous Materials controls, Laser Safety training, Organizational Conflict of Interest (OCI) training, CITI training for Protection of Human Subjects, Occupational Health training for animal use protocols, Export Control training, 711 HPW Biosafety Program training, and other mandatory military installation training requirements. Safety violations may require additional training and a medical evaluation.

7.0 Deliverables: For each research area, the Contractor shall be responsible for generating deliverables as defined in the overall contract data requirements list (CDRL) and specific data item descriptions; examples of deliverables include but are not limited to journal publications, technical reports, oral presentations, posters, software, and hardware. These deliverables may be developed for a broad and varied audience, including stakeholders in the DoD, source developers, safety standard committees, industry, and academia.

7.1. Data: It shall be the responsibility of the Contractor through the Program Manager and Subtask Leader to meet all reporting and documentation requirements of the ORBS contract. Reporting shall, at a minimum, include delivery of a monthly status report, and a monthly financial report highlighting financial activity to the subtask level. Quarterly summary data shall also be provided via a quarterly status report summarizing technical activity, and via a quarterly Contract Funds Status Report. In addition there are a number of data items that shall be required on an "as required" basis, such as Scientific and Technical Reports, to document a final report; and Technical Report-Study/Services to document research results on subtask (projects) in either AFRL Technical Report format or Scientific Journal Article format; and to document hardware and/or software developed as a part of this research. Also, slides for the annual Program Management Review on the overall contract shall be delivered per Presentation Material.

7.2. Hardware: Hardware that has been developed as a part of this program shall be delivered to the Government at the end of the contract period, or earlier if desired by the Government. The Contractor shall prepare such hardware items for delivery to the Government and shall prepare the appropriate data as specified in the contract CDRL.

7.3. Software: The Contractor shall deliver executable software developed under this contract along with approved documentation at the end of the contract period, or earlier if desired by the Government. The Contractor shall prepare such software items for delivery to the Government and shall prepare the appropriate data as specified in the contract CDRL.

8.0 Security:

- a. Individual Clearances. Upon receipt of a final DD Form 254, the Contractor shall obtain a U.S. security clearance at the minimum level of Secret for all contractor personnel required to have access

to classified information and/or require IT-II or IT-I level access. Onsite contractor personnel should have an active clearance prior to reporting for duty in support of any task order. Interim clearances for newly hired personnel shall be processed as expeditiously as possible since some contractor personnel will be required to utilize the SIPRNET to process classified materials; however this will be on a case-by-case basis. Such clearance must be obtained through the Defense Investigative Services. As required, for the performance of the contract the Contractor shall ensure those contractors requiring access to TS/SCI, and /or the individual contractor requires access to JWICS, have the required, favorably adjudicated TS investigation and have been granted SCI eligibility. The Contractor shall ensure that employees meet the personnel security clearance requirements of the National Industrial Security Program (NISPO), Chapter 2, Section 2. The Contractor shall provide current listing of employees prior to the start of the contract and immediately upon any updates to an employee's status or information change. The list shall include employee's name, social security number, and level of security clearance for all contractors performing work at 711 HPW/RHD, Fort Sam Houston, TX. The Contractors Facility Security Officer (FSO) shall validate the list and provide to the Sponsoring Agency's Unit Security Manager via a Visit Request submitted in DISS. The 711 HPW, Ft Sam Houston SMO Code is: JBBSA-FSHPW.

b. Information Security and Force Protection. The contractor shall comply with the Information Security and Force Protection requirements as defined by DoDM 5200.01 Vol 1-3 (DoD Information Security), AFMAN16-1404 (Information Security Program Management), and DoDI 2000.16 V-1, AFI 10-245-O (AT Program Implementation). Contractors performing work at 711 HPW/RHD, Fort Sam Houston, TX shall participate in the 711 HPW sustained Information Security and Force Protection/Anti-terrorism training program as provided/required by the 711 HPW/RHD Security Managers and Unit Antiterrorism Representative (UATR). The 711 HPW/RHD Training Manager will evaluate the training posture of AF contract activities and operations. Participation in the unit's Information Security program will include compliance with the unit's established in and out-processing requirements. This requirement is set forth in the guidance above and applicable AFMC, AFRL, and local supplements.

c. Operations Security (OPSEC). The contractor shall provide OPSEC protection for all sensitive/critical information as defined by AFI 10-701 (Operations Security), the TSRL OPSEC Plan, and critical information list. The contractor shall participate in the 711 HPW/RHD sustained OPSEC awareness training program as part of their on-going security education and training. The 711 HPW/RHD OPSEC coordinator will evaluate the OPSEC posture of AF contract activities and operations.

d. Access Credentials. Contractors performing work at 711 HPW/RHD, Fort Sam Houston, TX Contractors will be issued facility access credentials upon in-processing. Contractors are required to display the access credentials when inside of the facility and immediately remove the credential upon departure from the facility. The contractors will be provided a unique PIN which is not to be shared with any other individual. Contractors are required to notify the 711 HPW/RHD Security Managers immediately in the event the access credential is lost or stolen. The Contractor shall obtain the pass and identification items required for contract performance. The Contractor shall obtain base identification badges from the appropriate security office for each Contractor employee. The Contractor shall obtain all other required access badges, such as computer facilities access badges, computer access ID numbers and passwords from the Government.

e. Physical Security. Contractors performing work at 711 HPW/RHD, Fort Sam Houston, TX shall be responsible for safeguarding all government equipment, information and property provided for contractor use. At the close of each work period, government facilities, equipment, and materials shall

be secured. When authorized in writing by the unit of assignment unit commander, contractors may be authorized to Open/Close the facility. Specific facility Opening/Closing training will be provided by 711 HPW/RHD Security Managers. The Contractor shall comply with established security procedures. Security support requiring joint AF and Contractor coordination includes, but is not limited to, packaging classified information, mailing and receiving classified materials, implementing emergency procedures for protection of classified information, security checks, and internal security controls for protection of classified material and high value pilferable property.

f. Contractor-Supplied Personal Electronic Devices (PED). Only validated contractor-supplied PEDs, which serve a justified mission requirement in support of this contract, will be considered and authorized by the appropriate Information Protection Office and/or Security Manager. All required contractor-supplied PEDs will be listed in the Statement of Work and identified in the Department of Defense Form 254, *Department of Defense Contract Security Classification Specification*, in Block 13. Device listing must include Type, Model, Serial Number, and justification for usage. Contractors utilizing contractor-supplied PEDs will be required to adhere to establish usage policies and procedures to include being issued a property pass to carry contractor-supplied PEDs in and out of facilities in support of an AF contract. *Exception:* Individuals issued a contractor-supplied fitness device must contact TSRL Security Managers prior to introducing the device into the secure areas of the facility.

g. Security Agreement for Integrated Visitor Groups. The contractor shall provide Acknowledgement of Receipt for a local specific Contractor Security Agreement to protect classified information involved in performance under this contract or Task Order. The Agreement will outline responsibilities to include the following areas: Contractor security supervision; Standard Practice Procedures; access, accountability, storage, and transmission of classified material; marking requirements; security education; personnel security clearances; reports; security checks; security guidance; emergency protection; protection of government resources; DD Forms 254; periodic security reviews; and other responsibilities, as required.

h. TEMPEST/INFOSEC/COMPUSEC. The Contractor shall implement TEMPEST, Communications Security, Information Security (INFOSEC), and Computer Security (COMPUSEC), measures IAW Government, host base, and assigned unit policies as required, to include participation in any/all training requirements for the above disciplines. The Contractor shall safeguard Government property and controlled forms provided for Contractor use. At the end of each work period and when authorized, the Contractor shall secure Government facilities, equipment, and materials.

i. Base Traffic Regulations. The Contractor shall comply with base regulations pertaining to the possession of weapons, firearms and ammunition, speed limits and use of cell phones.

9. Associate Contractor Agreements:

a. The Contractor shall enter into Associate Contractor Agreements (ACA) for any portion of the contract requiring joint participation in the accomplishment of the Government's requirement. The agreements shall include the basis for sharing information, data, technical knowledge, expertise, and/or resources essential to the integration of the Optical Radiation Bioeffects and Safety, which shall ensure the greatest degree of cooperation for the development of the program to meet the terms of the contract.

b. ACAs shall include the following general information:

(1) Identify the associate contractors and their relationships.

- (2) Identify the program involved and the relevant Government contracts of the associate contractors.
 - (3) Describe the associate contractor interfaces by general subject matter.
 - (4) Specify the categories of information to be exchanged or support to be provided.
 - (5) Include the expiration date (or event) of the ACA.
 - (6) Identify potential conflicts between relevant Government contracts and the ACA; include agreements on protection of proprietary data and restrictions on employees.
- c. A copy of such agreement shall be provided to the Contracting Officer for review before execution of the document by the cooperating contractors.
- d. The Contractor is not relieved of any contract requirements or entitled to any adjustments to the contract terms because of a failure to resolve a disagreement with an associate contractor.
- e. Liability for the improper disclosure of any proprietary data contained in or referenced by any agreement shall rest with the parties to the agreement, and not the Government.
- f. All costs associated with the agreements are included in the negotiated cost of this contract. Agreements may be amended as required by the Government during the performance of this contract.